

MinION Loading

How much concentration do I really need?

- Final pool
 - Need 100-150 fmol to load the flow cell
 - = 11ul of 10-15nM
- Individual samples:
 - Ideally individual samples would also have 10-15nM
 - (7-10ng/ul for 1000bp library)
 - (13-20ng/ul for 2000bp library)
 - Total mass (ng) per sample added to pool will depend on:
 - library size
 - # of samples in the pool

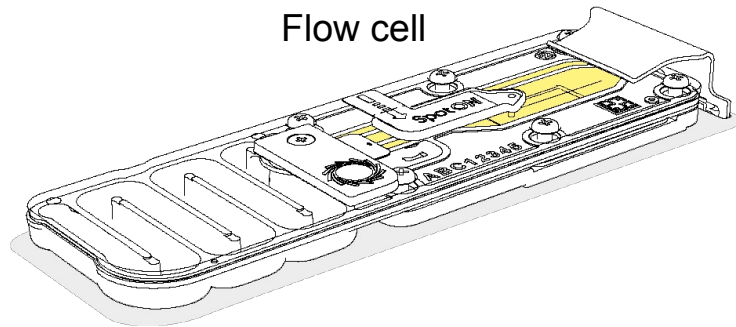
Before Loading...

- Thaw loading reagents
 - ~30 min at room temperature
- Take out SpotON flow cell from 4°C
- If it's your first time loading, watch [this video](#) to see how it is done
- Ensure that the MinION is connected to the computer and the software (MinKNOW) is up to date
- Insert the flow-cell in the instrument
- Perform a flow cell check and ensure that at least 800 pores are available* (ideally 1200-1800)

MinION instrument

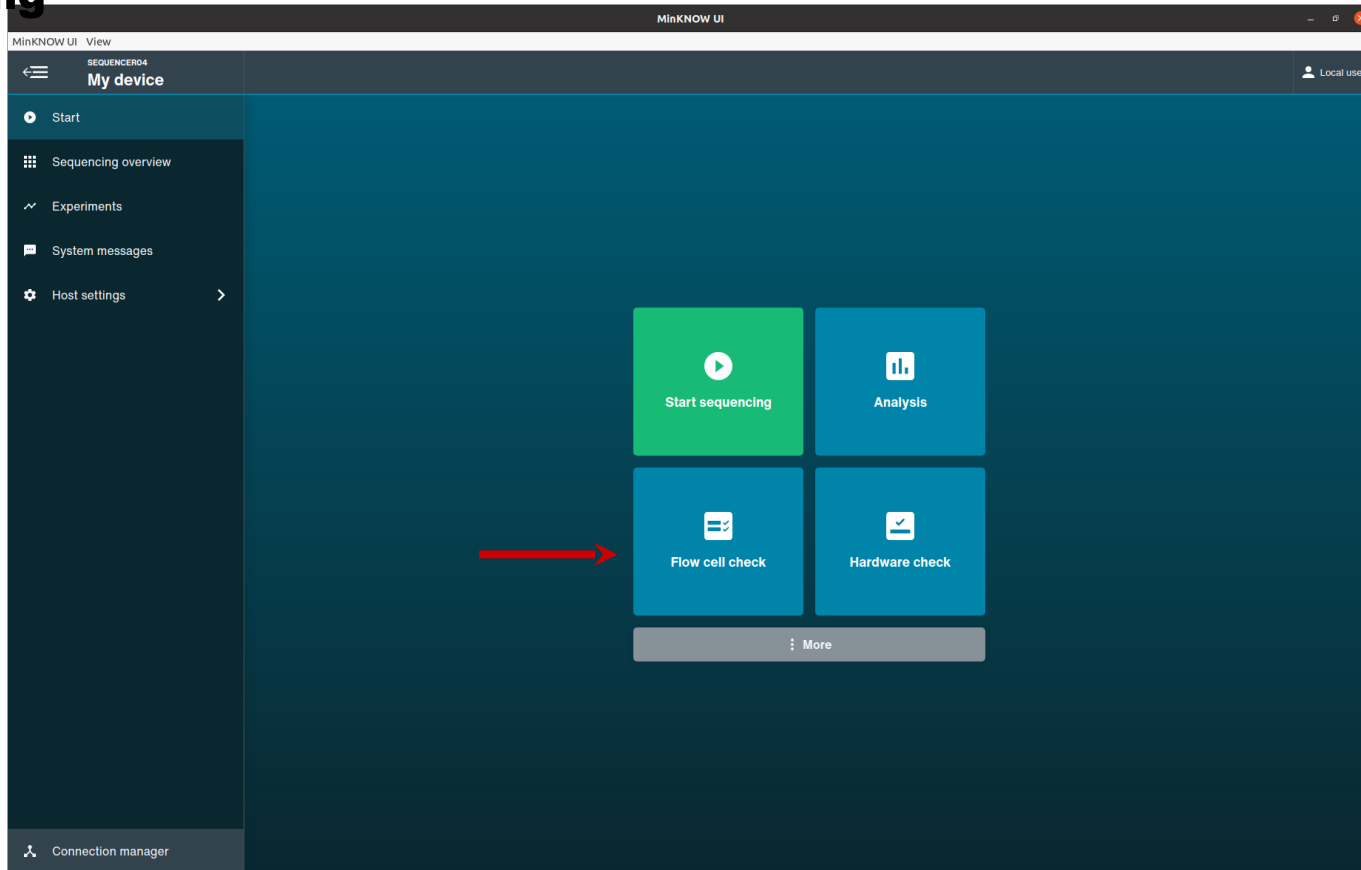


Flow cell



*ONT guarantees 800 active pores; you can sequence with fewer, but less data will be generated

Perform a Flow cell check to identify the number of available pores for sequencing



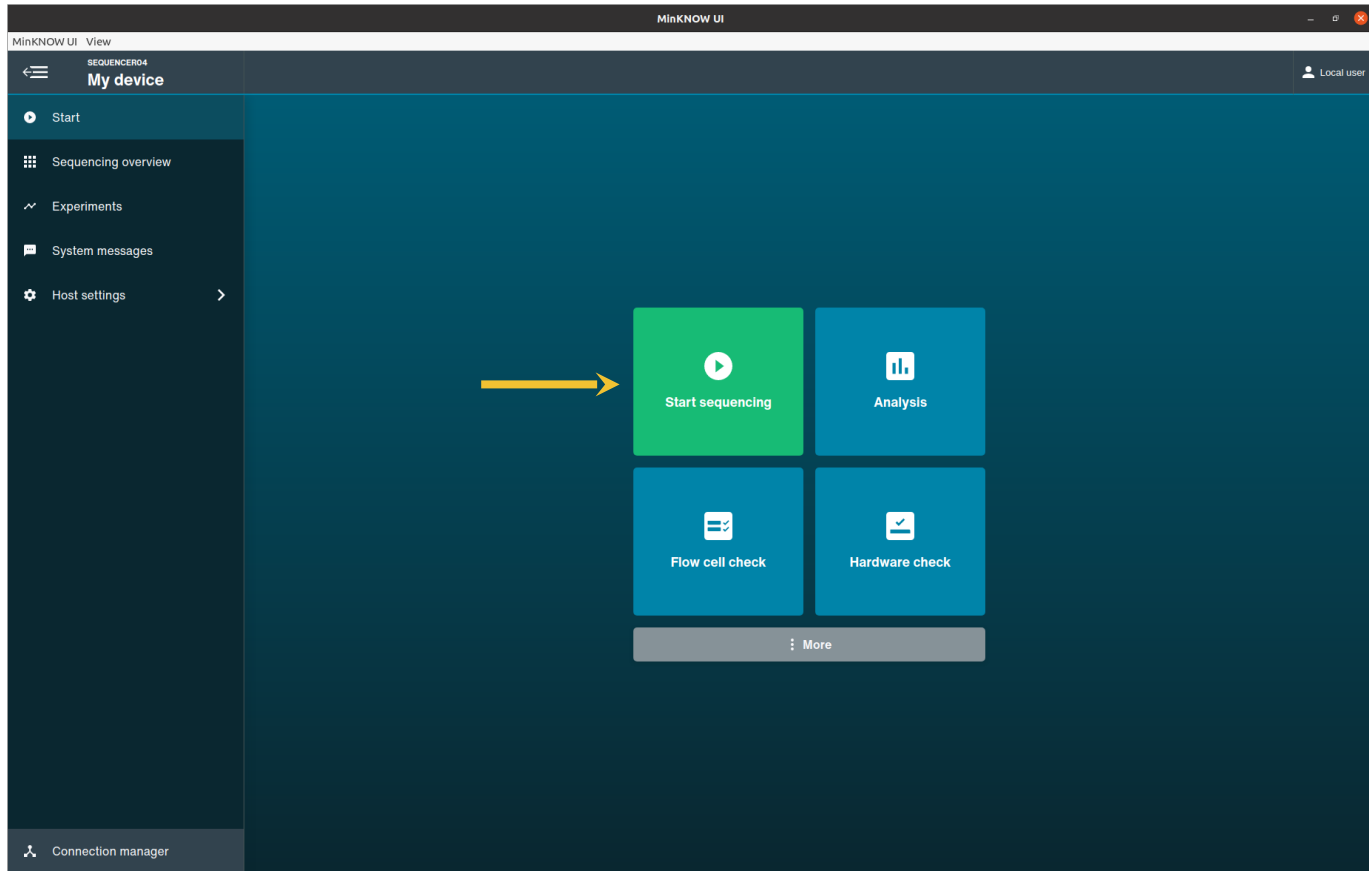
Loading the flow cell

- Follow the ONT protocol for “Priming and loading the MinION and GridION Flow Cell” (link [here](#))
- If it is your first time loading, watch the “Priming and loading your flow cell” video (link [here](#))

Notes:

- Generally, library beads (LIB) are used for loading, and library solution (LIS) is not used
- Use Bovine Serum Albumin (BSA) if you have the exact catalog number listed in the protocol (otherwise skip)

Starting a sequencing run



Starting a sequencing run

SFO-K4L90W06WC

My device

Start

Sequencing overview

Experiments

System messages

Host settings

Connection manager

MinKNOW UI

Help

G

1. Positions

2. Kit

3. Run configuration

Positions

Experiment details

Experiment name *
20250130_RR177_ONT_Pool1_150fmol

Join existing

Device positions

Position	Flow cell ID	Flow cell type	Sample ID
MN36411	FBA46076	FLO-MIN114	Pool1

< Back

Continue >

Select library prep kit used: SQK-RPB114-24 Rapid PCR Barcoding Kit

SFO-K4L90W06WC

My device

Start

Sequencing overview

Experiments

System messages

Host settings

Connection manager

MinKNOW UI

Help

G

1. Positions

2. Kit

3. Run configuration

Kit selection

Sample type

DNA

RNA

PCR-free

PCR

PCR-free

Multiplexing

Yes

No

Control

Reset filters

Ligation Sequencing Kit SQK-LSK114	Ligation Sequencing Kit SQK-LSK114-XL	Rapid Barcoding Kit 96 SQK-RBK114-96	Native Barcoding Sequencing Kit 24 SQK-NBD114-24
Native Barcoding Sequencing Kit 96 SQK-NBD114-96	Ultra-Long DNA Sequencing Kit SQK-ULK114	16S Barcoding Kit SQK-16S114-24	Multiplex Ligation Sequencing Kit XL SQK-MLK114-96-XL
PCR cDNA Barcoding Kit - 24 SQK-PCB114-24	PCR-cDNA Sequencing Kit SQK-PCS114	Rapid Sequencing Kit SQK-RAD114	Rapid Barcoding Kit 24 SQK-RBK114-24
Rapid PCR Barcoding Kit SQK-RPB114-24			

Select appropriate kit used for library prep

Back

Continue

Select a basecaller

MinKNOW UI

SFO-K4L9OW06WC

My device

Help

G

1. Positions

2. Kit

3. Run configuration

Run configuration

Load configuration

Experiment name	20250130_RR177_ONT_Pool1_150fmol
Sample ID	RR177_Pool1
Flow cell type	FLO-MIN114
Sequencing kit	Rapid PCR Barcoding Kit (SQK-RPB114-24)

Sequencing and analysis

Basecalling ☒

Modified bases Off

Duplex Off

Model High-accuracy basecalling

Barcoding ☒

Trim On

Barcode both ends Off

Selected barcodes All

Alignment ☐

Select appropriate basecaller

< Back

Save configuration

Start >>

Connection manager

Select a basecaller

SFO-K4L9OW06WC

My device

Start

Sequencing overview

Experiments

System messages

Host settings

Connection manager

1. Positions

2. Kit

3. Run configuration

Run configuration

Load configuration

Experiment name

Sample ID

Flow cell type

Sequencing kit

20250130_RR177_ONT_Pool1_150fmol

RR177_Pool1

FLO-MIN114

Rapid PCR Barcoding Kit (SQK-RPB114-24)

Sequencing and analysis

Basecalling

Modified bases

Duplex

Model

High-accuracy basecalling

Barcoding

Trim

Barcode both ends

Selected barcodes

Alignment

20250130_RR177_ONT_Pool1_150fmol

RR177_Pool1

FLO-MIN114

Rapid PCR Barcoding Kit (SQK-RPB114-24)

On

Off

All

Off

Off

High-accuracy basecalling

On

Off

All

Off

High-accuracy or super-accurate basecalling is recommended

(the choice is a balance between accuracy and available computational power)

Select appropriate basecaller

Back

Save configuration

Start

Intro to basecallers

What do they do?

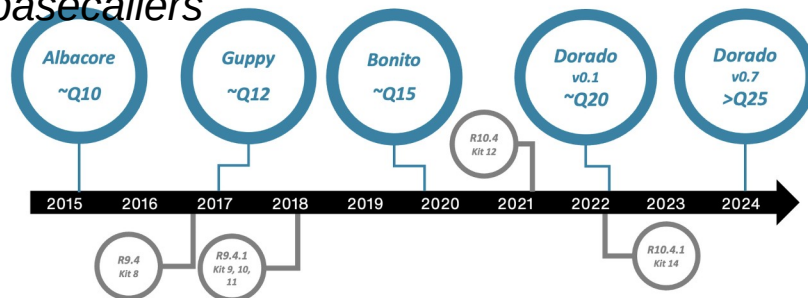
- Input raw ONT data (POD5)
- Output basecalled data (FASTQ)

Basecaller options

- Fast
- High Accuracy (HAC)
- Super Accuracy (SUP)

ONT Basecaller programs

ONT periodically releases new basecallers



Basecallers differ in accuracy, computational power required, and time



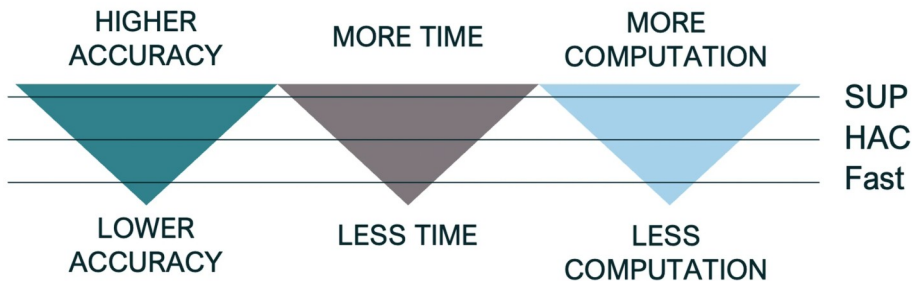
Experiment run time

3CGCTCGCTGATAGTCGTAGATGAATGCGCTCGCTGATAGTCGTAGAT

MinKNOW basecalling during experiment

MinKNOW continues basecalling

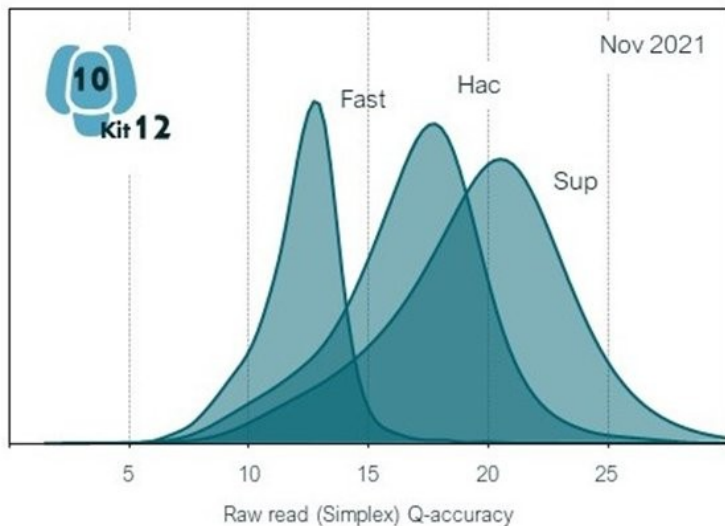
FASTQ/
BAM



Higher accuracy basecallers will require more computation and time

Fast basecallers can keep up with sequencing runs

Comparison of Dorado basecalling models - accuracy



Basecalling accuracy is measured in Phred quality score (Q score)

A Phred quality score (Q score) is a measure of confidence of the basecall based on the estimated error rate.

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10,000	99.99%

The choice of basecaller is dependent on the computational resources available, time, and accuracy required.

Fast basecalling: fastest, least computationally intense. Compatible with real-time basecalling on all nanopore devices with compute. Recommended for quick, real-time insights on sequencing data when compute resources are limited.

High accuracy basecalling (HAC): highly accurate, intermediate speed and computational requirement. Recommended for high-throughput projects focusing on variant analysis.

Super accuracy basecalling (SUP): the most accurate and computationally intense. Recommended for de novo assembly projects and low-frequency variant analysis (e.g. somatic variation and single-cell applications).

So which basecaller should I use?

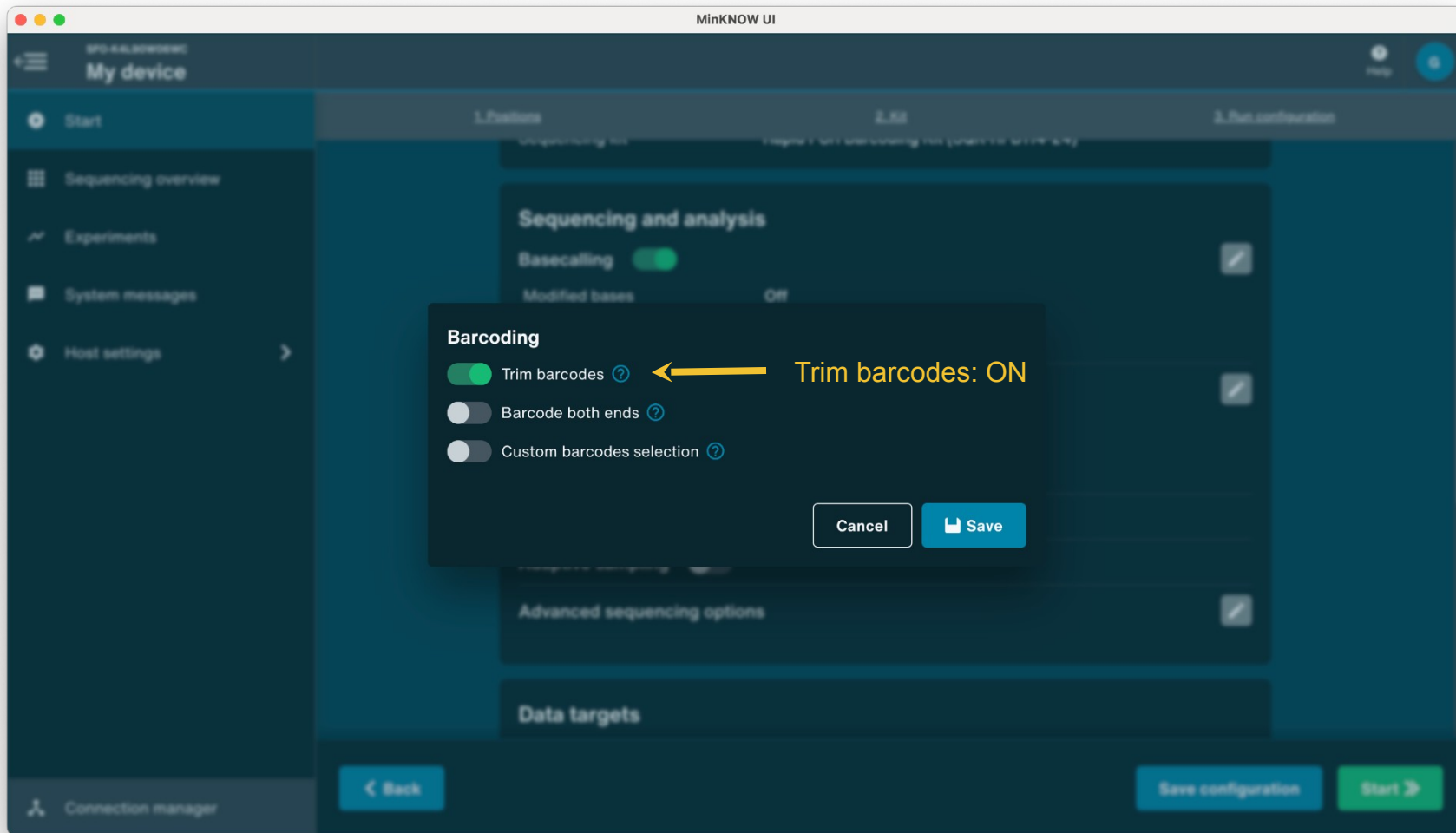
Real time during the run: HAC if possible, but if computation is limited, choose fast

Preliminary data analysis: fast is okay

Final data analysis: HAC or SUP* is recommended

**SUP may be too computationally intensive*

Trim barcodes: ON



Set run duration

MinKNOW UI

My device

Start

Sequencing overview

Experiments

System messages

Host settings

1. Position

2. Kit

3. Run configuration

Run targets

Run limit ⓘ

Action	Condition	Value
Stop run when	Time equals ▼	24 Hrs

Must be between 0.05—1,000

Flow cell data target (optional) ⓘ

Action	Condition	Value
Stop run when	Select ▼	Input Gb ⓘ

Maximum run time is 72 hrs
(data will be generated in near real-time regardless)

Pores get exhausted over time, so increasing run time may have diminishing returns

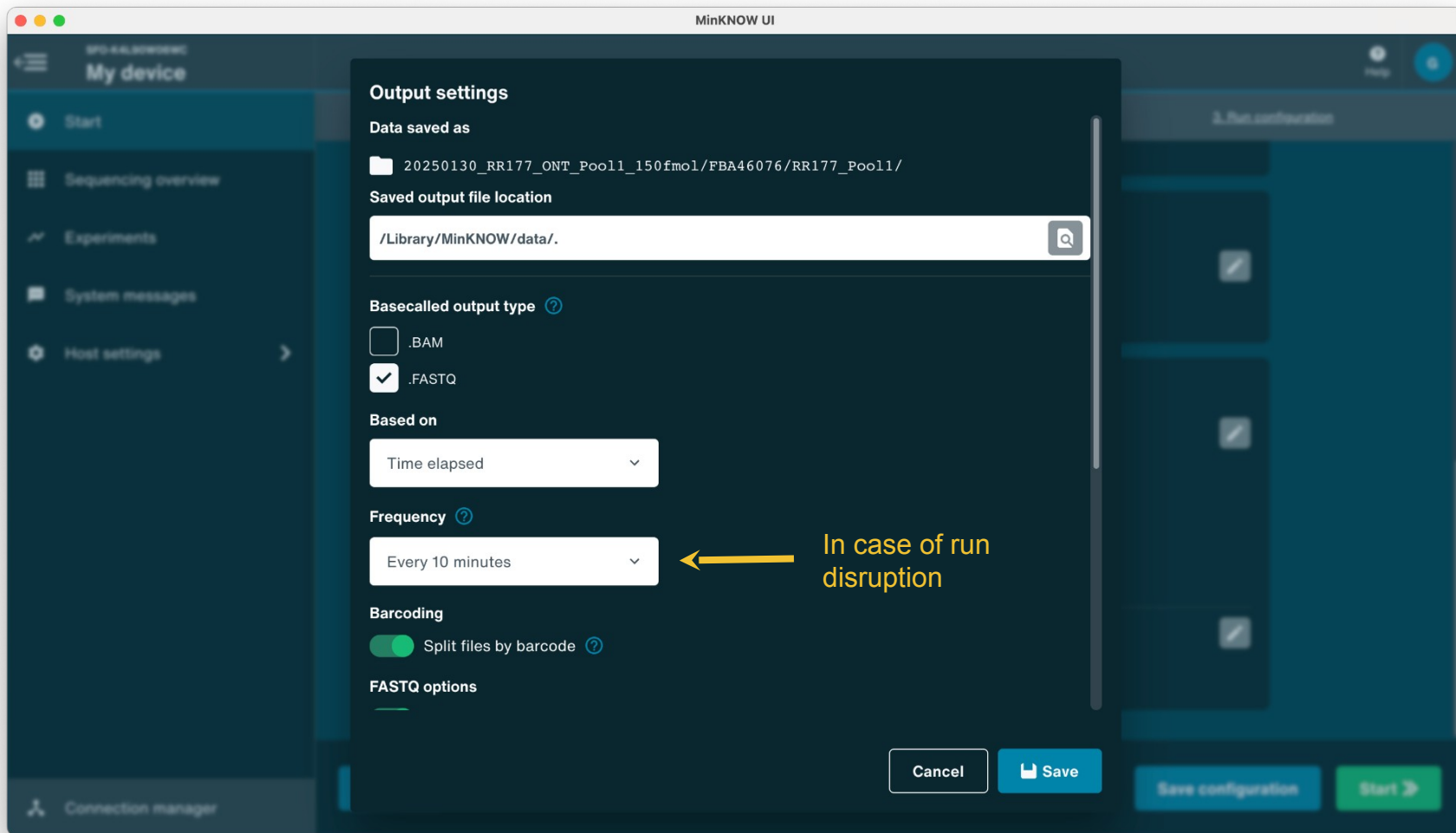
Cancel Save

← Back

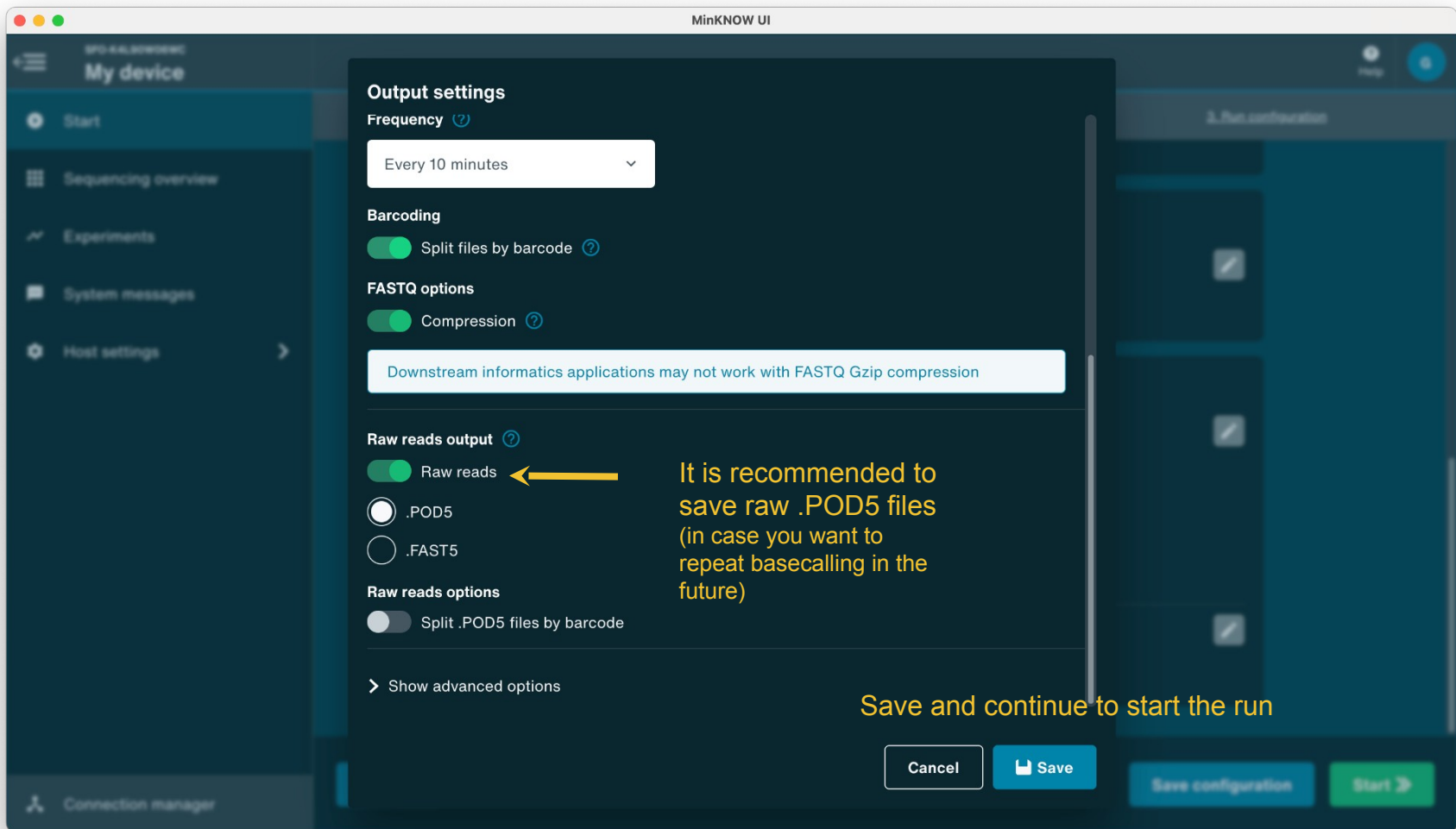
Save configuration Start →

Set run duration

Save output files every 10 minutes



Choose output files to store



Start the run

MinKNOW UI · View

SEQUENCER04
My device

Local user

Start

Sequencing overview

Experiments

System messages

Host settings

Connection manager

Sequencing overview

MinION Mk1B · MN28928 (Removable device) | PromethION · P2S-01827

MinION Mk1B · MN28928 (Removable device)

MN28928

FAY32108

Getting to 35°C

P2S-01827-A PAY88939

Ready for sequencing

Run complete

P2S-01827-B PAY89677

Ready for sequencing

Run complete

Flow cell warms up to 37°C and another flow cell check is performed

Then run will begin

Monitor the run

SFO-K4L90W06WC
My device

Start

Sequencing overview

Experiments

System messages

Host settings >

Connection manager

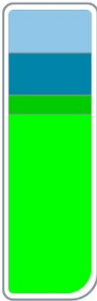
MinKNOW UI

Help G

Sequencing overview

MinION Mk1B · MN36411 (Removable device)

MN36411 FBA46076



Sequencing

Click on flow cell image to monitor run

Monitor the run

MinKNOW UI

MinKNOW UI View

SEQUENCER04

My device

Local user

Start

Sequencing overview

Experiments

System messages

Host settings

Connection manager

RR182_20240813_Test

Active

Experiment summary

Reads 3.06 k

Estimated / basecalled bases 2.16 Mb / 1.5 Mb

Active runs 1

Total runs 1

Run controls

Start pore scan

Export run report

Position

Flow cell ID

Sample ID

Health

Available pores

Run time

Run state

Reads

Estimated bases

Basecalled bases

Basecalled %

Started

MN28928	FAY32108	OC43		898 / 2048	5m 56s / 1d 0h	Active	3.06 k	2.16 Mb	1.5 Mb	71%	13 August 202
---------	----------	------	-------------	-----------------------	----------------	--------	--------	---------	--------	-----	---------------

MinION Mk1B · MN28928

Experiment group

RR182_20240813_Test

Sample ID

OC43

Flow cell product code

FLO-MIN114

Kit ID

SQK-RPB114-24

Current output directory

/mnt/ess_nanopore/raw/20240813_Gowtham/RR182_20240813_Test/OC43/20240813_1253...

Basecall model

High-accuracy basecalling

Minimum qscore

9

Total run time

5m 56s

Remaining time

1d 0h 1m

Temperature

35 °C

Read count

3.06 k

Estimated bases

2.16 Mb

Approx. N50

792.77 b

Basecalled bases

1.5 Mb

After a run

1. Check the run report to see approximately how many pores are available
1. Wash the flow cell if it is worth saving
1. Run flow cell check to see how many pores are still available after wash
1. Start another run *or* add storage buffer and save at 4C

Computer considerations

- POD5 files are very large
 - Ensure there is a storage plan in place to maintain space on the computer
 - You may want to move or discard POD5 files at a regular cadence
- Always have 300GB available before every run
 - A backup 5TB+ drive or a NAS (Network attached storage) are options to use for longer storage to keep your local storage clean